

CLOUD COMPUTING AND BIG DATA ANALYTICS

CSA15 – Assignment Answer

Design and Deployment of a Cloud-Based Virtualized Collaboration Platform for an Engineering College

Student Details	Value
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COURSE CODE	CSA1507
COURSE NAME	Cloud Computing and Big Data Analytics
CO	CO1, CO2, CO3
SDG Mapping	SDG 4 – Quality Education; SDG 8 – Decent Work and Economic Growth; SDG 9 – Industry, Innovation and Infrastructure

1. Problem Analysis

A small engineering college currently operates separate physical systems for application hosting, file storage, and user access. This leads to high infrastructure cost, inefficient resource utilization, difficult capacity management during peak demand, and additional administrative overhead. The proposed solution is a cloud-based virtualized environment that combines infrastructure virtualization, cloud service models, secure remote access, and collaboration services.

1.1 Institutional Requirements

Requirement	Proposed Cloud/Virtualized Solution	Reason
Online learning	LMS/application VM with web access	Supports faculty and students through browsers and remote access.
File sharing	Dedicated file-storage VM / cloud storage	Centralized storage with controlled access.
Application hosting	Application server VM / PaaS	Separates application deployment from physical hardware.
User authentication	Identity/access service	Provides controlled access based on user roles.
Collaborative editing	SaaS collaboration suite	Enables simultaneous document editing and sharing.
Communication	Cloud email/chat/video-conferencing service	Supports academic and administrative communication.
Scalability	Virtual CPU/RAM/storage allocation and cloud scaling	Resources can be increased when demand rises.
Security	Firewall, IAM, least privilege, encryption, backups	Protects institutional data and services.

2. Cloud Service Model Identification and Requirement Analysis

The proposed platform uses IaaS, PaaS, and SaaS together. Each model is assigned to the type of responsibility that it handles most effectively.

Service Model	Example in Proposed Platform	Institution Responsibility	Main Use
IaaS	Virtual machines, virtual network, storage, firewall	Configure OS, VM resources, applications and access policies	Infrastructure services and VM-based data center
PaaS	Managed application/database/runtime platform	Develop, configure and maintain application code	Application deployment without managing underlying servers
SaaS	Cloud office suite, online storage, communication and collaboration tools	Manage users, permissions and content	Document sharing, editing, communication and productivity

2.1 Analysis of IaaS

- IaaS provides virtualized compute, storage and networking resources while allowing the institution to control the operating systems and deployed services.
- It is suitable for the virtualized data-center demonstration because separate VMs can represent the LMS, file server, application server and monitoring/administration services.
- Scalability is achieved by changing VM CPU, memory and storage allocations or creating additional instances.
- The institution pays for allocated/consumed infrastructure rather than purchasing a separate physical server for every service.
- Security responsibilities remain significant because the institution must configure operating-system security, user permissions, firewall rules, patching and application security.

2.2 Analysis of PaaS

- PaaS provides a managed application runtime, reducing the need to administer the underlying operating system and server infrastructure.
- It is appropriate for college-developed portals, attendance applications, departmental applications and APIs.
- PaaS improves developer productivity because deployment, runtime management and scaling are handled by the platform.
- The major limitation is reduced infrastructure-level control and possible vendor/platform dependency.

- Security still requires secure application code, identity management, secrets management and correct configuration.

2.3 Analysis of SaaS

- SaaS delivers complete applications through the network and is the most appropriate model for collaboration activities.
- Examples include browser-based document editing, file sharing, email, chat, calendars and online meetings.
- SaaS minimizes installation and maintenance work for the college IT team.
- Accessibility is high because users can work from campus or remotely using supported devices.
- The main limitations are dependency on the service provider, network availability, subscription cost and reduced control over the underlying application platform.

2.4 Overall Service Model Comparison

Aspect	IaaS	PaaS	SaaS
Control	High	Medium	Low
Administration effort	High	Medium	Low
Scalability	High	High	Provider-managed
Typical college use	VMs, storage, networks	Application development	Collaboration/productivity
Cost model	Resource-based	Platform/usage-based	Subscription/usage-based
Security responsibility	Shared; institution manages more layers	Shared application/platform responsibility	Mostly provider-managed infrastructure; institution manages users/data

3. Virtualized Data Center Architecture

A small virtualized data center is proposed using a Type-2 virtualization environment such as VirtualBox for laboratory demonstration. The same logical design can be mapped to an IaaS provider for production use.

3.1 Logical Architecture

Layer / Component	Connectivity	Layer / Component
Students / Faculty / Admin	Internet / Campus Network	Secure Gateway / Firewall
	↓	
	Virtualization Host	

VM1 – LMS/Web Server	VM2 – File Server	VM3 – Application Server
VM4 – Database Server	VM5 Monitoring/Admin	Virtual Network / Internal LAN
	↓	
Cloud SaaS Collaboration: Documents + File Sharing + Communication		

3.2 VM Design

VM	Service	vCPU	RAM	Storage	Network
VM1	LMS / Web Server	2	4 GB	40 GB	Bridged Internal +
VM2	File Server	2	4 GB	100 GB	Bridged Internal +
VM3	Application Server	2	4 GB	50 GB	Internal
VM4	Database Server	2	4 GB	60 GB	Internal
VM5	Monitoring / Administration	1	2 GB	30 GB	Bridged Internal +

The exact allocation can be adjusted according to the host computer's available resources. For a classroom demonstration, the VMs can be started selectively rather than running every VM simultaneously.

4. Virtual Machine Deployment and Resource Configuration

The implementation can be performed in a controlled laboratory environment. The following procedure provides reproducible deployment steps and the evidence that should be captured in screenshots.

1. Install a virtualization tool such as VirtualBox on the host computer.
2. Create an internal virtual network for service-to-service communication.
3. Create VM1 for the LMS/Web service and install a Linux server operating system.
4. Create VM2 for file storage and configure a shared directory with user permissions.
5. Create VM3 for the college application service.

6. Create VM4 for the database service and restrict database access to the application network.
7. Create VM5 for administration/monitoring and secure remote management.
8. Assign CPU, memory, disk and network resources according to the VM design table.
9. Configure static or reserved IP addresses for infrastructure services.
10. Configure firewall rules so only required ports are exposed.
11. Test service-to-service communication using ping, SSH and application/database connection tests.
12. Capture screenshots of VM creation, CPU/RAM settings, storage settings, network settings and successful service access.

4.1 Example Resource Configuration

Host Network: Campus/Internet connection

Internal Network: CloudLab-Internal

VM1 IP: 192.168.50.11 -> Web/LMS

VM2 IP: 192.168.50.12 -> File Server

VM3 IP: 192.168.50.13 -> Application

VM4 IP: 192.168.50.14 -> Database

VM5 IP: 192.168.50.15 -> Administration

5. Network Configuration and Service Communication

The network is divided logically so that public-facing services and internal services do not have identical exposure. The web/LMS endpoint can be reachable from users, while the database and internal file services should remain restricted.

Source	Destination	Purpose	Security Rule
Student/Faculty	VM1	Access LMS/Web	Allow HTTPS; redirect HTTP to HTTPS where appropriate
Student/Faculty	VM2	Access permitted files	Allow only authenticated file access
VM3	VM4	Application database access	Allow database port only from VM3
Admin	VM1–VM5	Remote administration	Allow SSH/RDP only to authorized administrators
All VMs	Monitoring VM	Logs/monitoring	Allow required monitoring traffic

5.1 Connectivity Test

1. From VM3, ping VM4.
2. From VM3, test the database service on VM4.
3. From a client system, open the LMS URL hosted by VM1.
4. From an authorized account, access the shared files on VM2.
5. Attempt an unauthorized database connection from a student/client machine.
6. Confirm that the firewall blocks the unauthorized request.

6. Cloud Access and Remote Resource Management

Secure remote access is implemented using authenticated administrative access. SSH can be used for Linux VM administration, while browser-based HTTPS is used for user-facing applications. Administrative access should be restricted by firewall rules and strong authentication.

- Use HTTPS for browser-based services to protect credentials and data in transit.
- Use SSH keys rather than password-only administration where supported.
- Restrict management ports to trusted administrator networks or a VPN.
- Use multi-factor authentication for cloud/SaaS accounts whenever available.
- Maintain audit logs for authentication, file access and administrative actions.
- Back up critical configuration and academic data regularly.

7. User Roles, Permissions and Security

Role	Permissions	Restrictions
Administrator	Manage VMs, network, users, backups and security policies	Administrative access only for authorized IT staff
Faculty	Create/edit course materials, upload files, collaborate	Cannot change infrastructure or other users' permissions
Student	View/download permitted material, submit assignments, collaborate	Read/submit access; no administrative privileges
Guest	Limited public information	No access to private academic files or internal services

7.1 Security Controls

- Least-privilege access: users receive only the permissions required for their academic role.

- Network segmentation: database and management services remain on the internal network.
- Encryption: HTTPS/TLS protects web traffic; encrypted storage/backups should be used for sensitive data.
- Authentication: unique accounts, strong passwords and MFA for privileged/cloud accounts.
- Backup and recovery: scheduled backups protect against accidental deletion and service failure.
- Patching: operating systems and applications must be updated regularly.
- Monitoring: logs should be reviewed for failed logins, unusual access and service failures.

8. Cloud-Based Collaboration Services

Activity	Cloud Collaboration Function	Expected Benefit
Document sharing	Central cloud folder with role-based access	Single source of truth for academic documents
Collaborative editing	Multiple users edit shared documents	Reduces duplicate document versions
File access	Browser/mobile/desktop access to authorized files	Improves accessibility
Communication	Email, chat and online meetings	Supports remote academic coordination
Assignment submission	Controlled student submission folders	Simplifies collection and review
Department collaboration	Shared calendars and team workspaces	Improves coordination and scheduling

9. Comparison with Traditional Infrastructure

Aspect	Traditional Physical Infrastructure	Proposed Cloud/Virtualized Infrastructure
Resource utilization	Often low because servers are dedicated to individual services	Higher through VM consolidation and shared resources
Scalability	Requires purchasing/installing additional hardware	Resources can be resized or additional instances created
Accessibility	Often dependent on campus network/VPN setup	Services can be securely accessed remotely
Cost	High upfront hardware and	Lower upfront cost; recurring

	maintenance cost	resource/service cost
Management	Physical hardware and OS management	Centralized virtualization and cloud management
Security	Institution controls physical environment but must manage all security layers	Shared-responsibility model; requires correct IAM and configuration
Collaboration	File copies and local systems can create version conflicts	Centralized SaaS collaboration supports simultaneous work
Availability	Hardware failure may interrupt a service	Virtualization, backups and provider capabilities can improve resilience

9.1 Advantages

- Better resource utilization through virtualization.
- Rapid provisioning of new services.
- Improved scalability during examination or admission peaks.
- Remote access supports online learning and hybrid academic work.
- Centralized collaboration reduces duplicate files and version conflicts.
- Infrastructure can be managed from a smaller number of administrative interfaces.

9.2 Limitations and Challenges

- Dependence on reliable network connectivity.
- Cloud subscription and usage costs must be monitored.
- Misconfigured identity, storage or firewall rules can create security exposure.
- Vendor lock-in may make migration difficult.
- Virtualized environments require careful CPU, memory and storage planning.
- Large-scale cloud deployments require skilled administration and monitoring.

10. Implementation Results and Validation Plan

The following table defines the expected evidence for the completed laboratory implementation. Actual screenshots should be inserted after deployment; this report does not fabricate execution evidence.

CloudCollab

CAMPUS VDI 4.0

Virtualized Engineering Lab & Multi-Tenant Cloud Platform

Department: ALL CSE ECE MECH CIVIL BIOTECH

Aarav Sharma

STUDENT

22CS014

Virtual Workstations4 Active

Architecture & Design

IaC & Deployment

Faculty Lab Command

Capstone Projects

Cluster Telemetry

Automated Deployment Engine

Infrastructure-as-Code (IaC) CI/CD Orchestrator

One-Click Campus Cloud Provisioning & IaC Blueprint Vault

Turnkey deployment pipeline automating the end-to-end lifecycle of the virtualized engineering college platform. Combines modular Terraform scripts, Ansible playbooks, and Kubernetes Helm charts into a reproducible cluster rollout.

Reset Runner

Initiate Cloud Deployment

Target Environment

Campus On-Prem (OpenStack)

Worker Node Pool

16 Nodes

GPU Slices (A100/L4)

8 GPUs

Ceph NVMe Pool

120 TB

Idle VM Hibernation

30 Mins

Deployment Pipeline Stages

Idle

1 IaC VPC & Subnet Provisioning

Terraform v1.8.4

Creates isolated department subnets (VLAN 120-180) and NAT gateways.

2 Bare-Metal & Hypervisor Setup

Ansible Playbook

Configures KVM, libvirt, NVIDIA Container Runtime, and SSSD LDAP.

3 Ceph Storage Mesh & CSI

Rook-Ceph Operator

Initializes 120TB NVMe storage pool and persistent /home volume claims.

Live Orchestrator Console

Stream Connected

> Live Orchestrator Console

System ready for automated orchestration.

Click "Initiate Cloud Deployment" to trigger the IaC runner pipeline.

Virtual Workstations4 Active

Architecture & Design

IaC & Deployment

Faculty Lab Command

Capstone Projects

Cluster Telemetry

VDI

ubuntu-24-ml-ws-014

Live Session Connected

IP: 10.142.12.45 • Ubuntu 24.04 LTS (x86_64) • Hardware: 8 vCPU / 32GB RAM / NVIDIA RTX 4090 24GB

MutedCast ScreenAsk Faculty

Interactive Bash Shell

Cloud IDE & Compiler

3D CAD & FEA Stress Canvas

WebRTC 60 FPS

student@ubuntu-24-ws-014: ~

PTY Session #4812 (Active)

Welcome to Apex CloudCollab Virtualized Workstation v4.2

Connected to container: ubuntu-24-ml-ws-014 (10.142.12.45)

Type "help" to see available commands or "python3 main.py" to execute the ML model.

Lab Peers & Instructor Presence

3 Online

Aarav Sharma (You)

22CS014

OWNER

Rohan Gupta

22CS029 • Co-Typing Line 24

Dr. K. Ramanujan

Course Instructor (Viewing)

FACULTY

Session Team Chat

Encrypted Peer Socket

Dr. K. Ramanujan (Faculty)

02:15 PM

Remember to normalize input tensors before running the backup step for Question 3.

CloudCollab

CAMPUS VDI 4.0

Virtualized Engineering Lab & Multi-Tenant Cloud Platform

Department: ALL CSE ECE MECH CIVIL BIOTECH

Aarav Sharn

22CS014

Virtual Workstations4 Active

Architecture & Design

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Capstone Projects

Cluster Telemetry

Institutional Architecture Blueprint

Tier-3 Hybrid Campus Datacenter & Cloud Fabric

Cloud-Based Virtualized Collaboration Infrastructure Topology

High-availability multi-tenant cloud architecture engineered for modern engineering curricula. Enables zero-install browser VDI streaming, automated GPU slicing, department network isolation, and low-latency collaborative lab workstations across campus networks.

Hypervisor

KubeVirt + KVM

VDI Streaming

WebRTC HTML5

Shared Storage

120TB NVMe Ceph

TRAFFIC & VIRTUALIZATION FLOW PIPELINE

1 Zero-Trust Edge

Campus Firewall, 802.1X, SSO LDAP Auth, BGP Gateway

2 VDI Stream Broker

Traefik Ingress, Apache Guacamole, WebRTC proxy

3 Compute & GPU Mesh

KubeVirt VMs, Docker Pods, NVIDIA MIG GPU Slicing

4 Distributed Ceph

NVMe CephFS /home mounts, Read-Only Datasets

5 Snapshot Vault

Hourly delta snapshots, MinIO S3 backup, Git sync

CAMPUS VDI 4.0
Virtualized Engineering Lab & Multi-Tenant Cloud Platform

Department:

ALL
CSE
ECE
MECH
CIVIL
BIOTECH

Aarav Sharma
22CS014

Virtual Workstations
4 Active
Architecture & Design
IaC & Deployment
Faculty Lab Command
Capstone Projects
Cluster Telemetry

Faculty & Lab Director Portal
Curriculum Orchestration & Live Student Assist

Class Laboratory Session & Automated Evaluation Command Center

Monitor real-time student sandbox activity, remotely inspect stuck workspaces, trigger parallel automated unit test suites, and provision departmental laboratory clusters on demand.

Run Batch Auto-Grader
Provision Class Sandboxes

ACTIVE LABORATORY COURSES

CS302

Operating Systems & Virtualization Lab

60 Students

54 VMs Live

ACTIVE

EC401

Digital Signal Processing & FPGA Synthesis

45 Students

38 VMs Live

ACTIVE

ME204

Computer Aided Design & Thermal FEA Lab

40 Students

32 VMs Live

ACTIVE

CS420

Distributed Cloud Systems & Kubernetes...

50 Students

48 VMs Live

SCHEDULED

Live Student Workstations: CS302 - Operating Systems & Virtualization Lab

Instructor: Dr. K. Ramanujan | Auto-shutdown timer: 45m

2 Need Assistance

Instructor Remote Assist Terminal

014
Aarav Sharma
(22CS014)

VM: vm-cse-014 • Unit Tests: 19/20 Passed • Exec Time: 42ms

95 / 100
AUTO-GRADED
Takeover / Assist

No Student Selected

Virtual Workstations
4 Active
Architecture & Design
IaC & Deployment
Faculty Lab Command
Capstone Projects
Cluster Telemetry

Interdisciplinary Engineering Capstone Hub
Multi-Student Collaborative Workspaces

Shared Cloud Repositories & High-Performance Project Clusters

Equipped with CephFS shared cloud drives, pooled NVIDIA GPU nodes, automated git tracking, and cross-departmental team workspaces uniting CSE, ECE, Mechanical, and Civil engineering cohorts.

500 GB Shared NVMe

Autonomous Campus Delivery Rover (Project Garuda)
Lead: Aarav Sharma (22CS014) • 2 GPUs

Solar Microgrid & Energy Forecasting Node
Lead: Karan Patel (22EE018) • 1 GPU

Capstone Sprint Milestone Tasks
Click any task to advance its status to the next milestone

TO DO
1

IN PROGRESS
2

COMPLETED
1

Interdisciplinary Team Roster

Aarav Sharma
Perception & SLAM (CSE)
22CS014

Priya Nair
Motor Control & PCB Design (ECE)
22EC003

Siddharth V.
Chassis & Suspension FEA (MECH)
22ME007

Test	Expected Result	Evidence to Capture
VM creation	Five VMs visible in the virtualization manager	VM manager screenshot
Resource configuration	CPU/RAM/storage match the design	Settings screenshots
Network connectivity	Required VMs communicate successfully	Ping/network test screenshot
LMS/Web access	Authorized client opens the service	Browser screenshot
File access	Authorized users can access permitted files	File-service screenshot
Database security	Only application VM can access	Connection/firewall test

	database	
Remote administration	Authorized admin can securely manage VMs	SSH/VPN/admin screenshot
Role-based access	Student/faculty/admin permissions differ	Permission test screenshots
Collaboration	Users can share/edit documents according to permissions	Collaboration-service screenshot

11. Analysis of Scalability, Cost, Accessibility and Security

Factor	Analysis
Resource utilization	Virtual machines allow multiple services to share one physical host. CPU and memory can be allocated according to workload.
Scalability	Cloud resources can be resized or replicated as demand increases. This is useful during admissions, examinations and assignment deadlines.
Accessibility	Browser-based services and secure remote access allow students and faculty to use services beyond the physical campus.
Cost	Virtualization reduces the number of dedicated physical servers. Cloud services change capital expenditure into operating/resource-based expenditure, but ongoing usage must be controlled.
Security	Identity, least privilege, segmentation, encryption, patching, backups and monitoring are required. Cloud adoption does not remove security responsibility.
Collaboration	Centralized SaaS tools improve simultaneous editing, sharing and communication compared with manually exchanging file copies.

12. Deployment Algorithm / Pseudocode

CLOUD_VIRTUALIZED_COLLEGE_PLATFORM()

 analyze institutional requirements

 classify requirements as IaaS, PaaS or SaaS

 install virtualization platform

 create internal virtual network

- create VM1 for LMS/Web
- create VM2 for File Storage
- create VM3 for Application Service
- create VM4 for Database Service
- create VM5 for Administration/Monitoring

- assign CPU, RAM, storage and IP configuration
- install required operating systems and services
- configure firewall and network segmentation
- configure users and role-based permissions
- enable secure remote administration
- configure collaboration services

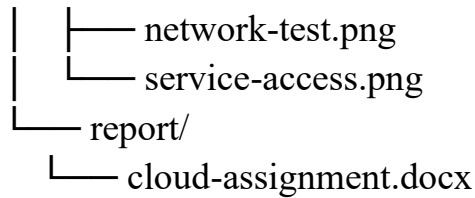
- test VM-to-VM communication
- test application, file and database access
- test unauthorized access restrictions
- configure backup and monitoring

- record screenshots and implementation results
- evaluate scalability, cost, accessibility, security and collaboration
- return final cloud platform design and results

13. GitHub Upload Structure

A clean repository can be organized as follows:

```
cloud-virtualized-collaboration-platform/  
├── README.md  
├── architecture/  
│   └── architecture-diagram.png  
├── virtualization/  
│   ├── vm-configuration.md  
│   └── network-configuration.md  
├── deployment/  
│   └── deployment-steps.md  
├── security/  
│   └── roles-and-permissions.md  
├── screenshots/  
│   └── vm-manager.png
```



The README should contain the problem statement, architecture, software requirements, deployment steps, test results, security considerations and limitations. Screenshots should be original evidence from the actual implementation.

14. Final Conclusion

The proposed cloud-based virtualized collaboration platform addresses the institution's requirements for online learning, file sharing, application hosting, secure access and collaborative academic activities. IaaS is appropriate for the virtualized infrastructure and infrastructure services, PaaS is suitable for managed application development and deployment, and SaaS is appropriate for collaboration and productivity.

The virtualized data-center design uses multiple VMs with dedicated responsibilities while sharing the underlying physical resources. Network segmentation and role-based access control reduce unnecessary exposure, while HTTPS, secure remote administration, backups, patching and monitoring provide essential security controls.

Compared with traditional physical infrastructure, the proposed design improves resource utilization, scalability, accessibility and collaboration, while introducing new considerations such as network dependency, recurring cloud cost, vendor dependency and shared-responsibility security. The design therefore provides a practical foundation for an engineering college to migrate services incrementally to a cloud environment.

15. Expected Learning Outcomes

- CO1: Identify and analyze IaaS, PaaS and SaaS according to institutional requirements.
- CO2: Design and configure a virtualized data-center environment with multiple service VMs and network communication.
- CO3: Demonstrate secure cloud access, role-based permissions and cloud collaboration mechanisms.
- Understand the trade-offs among resource utilization, scalability, cost, accessibility, security and collaboration.

16. Reference Basis

This answer follows the organization and expected deliverables of the provided DAA assignment reference: problem analysis, algorithms/procedures, step-by-step implementation, results/evidence, complexity or analytical discussion, comparison, limitations, and final conclusion. The Cloud Computing assignment's stated deliverables are architecture design, virtualized data-center implementation/results, cloud service model analysis, VM evidence, comparison/analysis, and GitHub upload.